JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Session : 2024-2025

Class Test -1 Examination

Branch: CSE/AI

Subject: Numerical Methods and Discrete Mathematics

Subject Code: CS/AI3M001

Date: 01/08/2024

Year/ Sem: - 2nd year/ III Sem

Marks: 20

Duration: 1 Hr.

Instructions:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to: (Listed Below CO & Mention it on question)

CO1: Student will be able describe the concept of Numerical methods, Bisection method and Newton-Raphson method and their convergence, Gauss Elimination method and Crout's method.

CO2: Student will be able to illustrate the concept of mathematical expectation, variance and standard deviation, moment and moment generating function, characteristic function, conditional expectation, variance and moment.

CO3: Student will be able to apply the concept of joint probability and random variable, probability function of CRV, joint probability function of DRV, joint distribution function of DRV.

CO4: Student will be able to Analyse the problem by using the concept of linear dependence and independence of vector, consistency of homogeneous and non-homogeneous equation.

CO5: Student will be able to Evaluate the problem based on the concept of eigen values and eigen vectors, Sylvester's theorem, Quadratic form, Canonical form, differentiation & integration of matrices.

CO6: Students will be able to create the new concept by using the theory of Group, group, Subgroups, homomorphism, cosets, Lagrange's theorem and Normal subgroup.

SECTION A

Q. No.	Answer in one word/sentence. (All Questions are compulsory).	Bloom's Level & CO	Marks
Q.1	Define Characteristic equation.	L1/CO1	1
Q.2	What is a consistent system of linear equation?	L1/CO1	1
Q.3	Outline the condition on $\propto_i$'s where i= 1,2, 3, …, n such that the set of vectors will be linearly independent.	L2/CO2	1
Q.4	Define Trivial solution.	L1/CO1	1

SECTION B

Q. No. **Solve answer in brief**

(All Questions are compulsory).

Q.5 Illustrate for what value of 'a' the system of linear equation has a unique solution **L2/CO2** **2**

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & a+3 \end{bmatrix}$$

Q.6 Interpret the quadratic form corresponding to the matrix **L2/CO2** **2**

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{bmatrix}$$

Q.7 Identify the rank, Index and Signature of the given matrix **L3/CO3** **2**

$$B = \begin{bmatrix} -\frac{5}{8} & 0 & 0 \\ 0 & \frac{6}{5} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Q.8 Solve the matrix given below and find it's characteristic equation **L3/CO3** **2**

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 2 & 3 \\ 4 & 5 & 8 \end{bmatrix}$$

SECTION C

Q. No. **Solve any two Questions**

Q.9 Test for consistency of the system **L4/CO4** **4**

$x + y + z = 3$
$x + 2y + 3z = 4$
$x + 4y + 9z = 6$
and solve if consistent.

Q.10 Simplify the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ by Sylvester's theorem and find A^{100}. **L4/CO4** **4**

Q.11 Identify the eigen vectors for the respective eigen values 2,3,6 for the matrix **L3/CO3** **4**

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$